MODUL 3

TUGAS PENDAHULUAN

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Tugas Pedahuluan :

1. table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.
   Script Table :

```oracle
CREATE TABLE mahasiswa (
    id           NUMBER PRIMARY KEY,
    nama         VARCHAR2(100),
    tempat_lahir VARCHAR2(50),
    tanggal_lahir DATE,
    nomor_hp     VARCHAR2(15),
    email        VARCHAR2(50),
    tinggi_badan NUMBER,
    berat_badan  NUMBER
);
```

   Screenshoots Table :

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID	NUMBER	No	(null)	1	(null)
2	NAMA	VARCHAR2(100 BYTE)	Yes	(null)	2	(null)
3	TEMPAT_LAHIR	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	TANGGAL_LAHIR	DATE	Yes	(null)	4	(null)
5	NOMOR_HP	VARCHAR2(15 BYTE)	Yes	(null)	5	(null)
6	EMAIL	VARCHAR2(50 BYTE)	Yes	(null)	6	(null)
7	TINGGI_BADAN	NUMBER	Yes	(null)	7	(null)
8	BERAT_BADAN	NUMBER	Yes	(null)	8	(null)

2. insert data pada table nomor 1 sebanyak 15 rows.
   Script Table :

TP_3 | MAHASISWA

0.066 seconds

Worksheet | Query Builder

```oracle
INSERT ALL
    INTO mahasiswa VALUES (1, 'Ulung Pratama', 'Purwokerto', TO_DATE('2004-05-15', 'YYYY-MM-DD'), '081234567890', 'ulung@email.com', 175, 72)
    INTO mahasiswa VALUES (2, 'Budi Santoso', 'Jakarta', TO_DATE('2003-11-20', 'YYYY-MM-DD'), '081298765432', 'budi.s@email.com', 170, 65)
    INTO mahasiswa VALUES (3, 'Siti Aminah', 'Bandung', TO_DATE('2005-01-10', 'YYYY-MM-DD'), '085712345678', 'siti_a@email.com', 160, 50)
    INTO mahasiswa VALUES (4, 'Andi Wijaya', 'Surabaya', TO_DATE('2004-08-25', 'YYYY-MM-DD'), '081344556677', 'andi.w@email.com', 180, 80)
    INTO mahasiswa VALUES (5, 'Dewi Lestari', 'Jogja', TO_DATE('2004-03-05', 'YYYY-MM-DD'), '089988776655', 'dewi.l@email.com', 158, 48)
    INTO mahasiswa VALUES (6, 'Fajar Ramadhan', 'Semarang', TO_DATE('2003-12-12', 'YYYY-MM-DD'), '082111223344', 'fajar_r@email.com', 172, 68)
    INTO mahasiswa VALUES (7, 'Gita Gutawa', 'Jakarta', TO_DATE('2005-06-18', 'YYYY-MM-DD'), '081122334455', 'gita.g@email.com', 163, 52)
    INTO mahasiswa VALUES (8, 'Hendra Gunawan', 'Medan', TO_DATE('2004-10-30', 'YYYY-MM-DD'), '085277889900', 'hendra_g@email.com', 178, 85)
    INTO mahasiswa VALUES (9, 'Indah Permata', 'Palembang', TO_DATE('2004-02-14', 'YYYY-MM-DD'), '087766554433', 'indah.p@email.com', 165, 55)
    INTO mahasiswa VALUES (10, 'Joko Widodo', 'Solo', TO_DATE('2003-09-09', 'YYYY-MM-DD'), '081200998877', 'jokowi@email.com', 170, 70)
    INTO mahasiswa VALUES (11, 'Kevin Sanjaya', 'Banyuwangi', TO_DATE('2004-07-07', 'YYYY-MM-DD'), '085811223344', 'kevin.s@email.com', 174, 66)
    INTO mahasiswa VALUES (12, 'Lesti Kejora', 'Cianjur', TO_DATE('2005-04-01', 'YYYY-MM-DD'), '089533445566', 'lesti.k@email.com', 155, 45)
    INTO mahasiswa VALUES (13, 'Maulana Malik', 'Makassar', TO_DATE('2004-11-11', 'YYYY-MM-DD'), '082212345678', 'maulana.m@email.com', 168, 62)
    INTO mahasiswa VALUES (14, 'Nina Bobo', 'Denpasar', TO_DATE('2005-02-28', 'YYYY-MM-DD'), '081999887766', 'nina.b@email.com', 160, 50)
    INTO mahasiswa VALUES (15, 'Oscar Pratama', 'Malang', TO_DATE('2003-05-05', 'YYYY-MM-DD'), '085311224455', 'oscar.p@email.com', 182, 88)
SELECT * FROM dual;
```

Screenshoot Table :

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NOMOR_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	1	Ulung Pratama	Purwokerto	15-05-2004	081234567890	ulung@email.com	175	72
2	2	Budi Santoso	Jakarta	20-11-2003	081298765432	budi.s@email.com	170	65
3	3	Siti Aminah	Bandung	10-01-2005	085712345678	siti_a@email.com	160	50
4	4	Andi Wijaya	Surabaya	25-08-2004	081344556677	andi.w@email.com	180	80
5	5	Dewi Lestari	Jogja	05-03-2004	089988776655	dewi.l@email.com	158	48
6	6	Fajar Ramadhan	Semarang	12-12-2003	082111223344	fajar_r@email.com	172	68
7	7	Gita Gutawa	Jakarta	18-06-2005	081122334455	gita.g@email.com	163	52
8	8	Hendra Gunawan	Medan	30-10-2004	085277889900	hendra_g@email.com	178	85
9	9	Indah Permata	Palembang	14-02-2004	087766554433	indah.p@email.com	165	55
10	10	Joko Widodo	Solo	09-09-2003	081200998877	jokowi@email.com	170	70
11	11	Kevin Sanjaya	Banyuwangi	07-07-2004	085811223344	kevin.s@email.com	174	66
12	12	Lesti Kejora	Cianjur	01-04-2005	089533445566	lesti.k@email.com	155	45
13	13	Maulana Malik	Makassar	11-11-2004	082212345678	maulana.m@email.com	168	62
14	14	Nina Bobo	Denpasar	28-02-2005	081999887766	nina.b@email.com	160	50
15	15	Oscar Pratama	Malang	05-05-2003	085311224455	oscar.p@email.com	182	88

3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:
   a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE):

Script Table :

TP_3 | MAHASISWA

Worksheet | Query Builder

```
SELECT
    LOWER(nama) AS nama_kecil,
    UPPER(nama) AS nama_kapital,
    INITCAP(tempat_lahir) AS tempat_rapi,
    CONCAT(nama, email) AS gabung_nama_email,
    LENGTH(nama) AS panjang_nama,
    SUBSTR(nama, 1, 5) AS lima_huruf_awal,
    LPAD(id, 5, '0') AS id_padding,
    RPAD(nomor_hp, 15, '*') AS hp_sensor,
    LTRIM('   data') AS hapus_kiri,
    RTRIM('data   ') AS hapus_kanan,
    INSTR(email, '@') AS posisi_at,
    REPLACE(email, 'email.com', 'mhs.univ.ac.id') AS email_baru
FROM mahasiswa WHERE id <= 15;
```

Screenshoot Output :

	NAMA_KECIL	NAMA_KAPITAL	TEMPAT_RAPI	GABUNG_NAMA_EMAIL	PANJANG_NAMA	LIMA_HURUF_AWAL	ID_PADDING	HP_SENSOR	HAPUS_KIRI	HAPUS_KANAN	POSISI_
1	ulung pratama	ULUNG PRATAMA	Purwokerto	Ulung Pratamaulung@email.com	13	Ulung	00001	081234567890***	data	data	
2	budi santoso	BUDI SANTOSO	Jakarta	Budi Santosobudi.s@email.com	12	Budi	00002	081298765432***	data	data	
3	siti aminah	SITI AMINAH	Bandung	Siti Aminahsiti_a@email.com	11	Siti	00003	085712345678***	data	data	
4	andi wijaya	ANDI WIJAYA	Surabaya	Andi Wijayaandi.w@email.com	11	Andi	00004	081344556677***	data	data	
5	dewi lestari	DEWI LESTARI	Jogja	Dewi Lestaridewi.l@email.com	12	Dewi	00005	089988776655***	data	data	
6	fajar ramadhan	FAJAR RAMADHAN	Semarang	Fajar Ramadhanfajar_r@email.com	14	Fajar	00006	082111223344***	data	data	
7	gita gutawa	GITA GUTAWA	Jakarta	Gita Gutawagita.g@email.com	11	Gita	00007	081122334455***	data	data	
8	hendra gunawan	HENDRA GUNAWAN	Medan	Hendra Gunawanhendra_g@email.com	14	Hendr	00008	085277889900***	data	data	
9	indah permata	INDAH PERMATA	Palembang	Indah Permataindah.p@email.com	13	Indah	00009	087766554433***	data	data	
10	joko widodo	JOKO WIDODO	Solo	Joko Widodojokowi@email.com	11	Joko	00010	081200998877***	data	data	
11	kevin sanjaya	KEVIN SANJAYA	Banyuwangi	Kevin Sanjayakevin.s@email.com	13	Kevin	00011	085811223344***	data	data	
12	lesti kejora	LESTI KEJORA	Cianjur	Lesti Kejoralesti.k@email.com	12	Lesti	00012	089533445566***	data	data	
13	maulana malik	MAULANA MALIK	Makassar	Maulana Malikmaulana.m@email.com	13	Maula	00013	082212345678***	data	data	
14	nina bobo	NINA BOBO	Denpasar	Nina Bobonina.b@email.com	9	Nina	00014	081999887766***	data	data	
15	oscar pratama	OSCAR PRATAMA	Malang	Oscar Pratamaoscar.p@email.com	13	Oscar	00015	085311224455***	data	data	

b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH| YEAR])) :

Script Table :

Worksheet | Query Builder

```
SELECT
    TO_CHAR(tanggal_lahir, 'DD / DY') AS format_hari,
    TO_CHAR(tanggal_lahir, 'YYYY / YY / RR') AS format_tahun,
    TO_CHAR(SYSDATE, 'HH24:MI:SS AM') AS jam_sekarang,
    ADD_MONTHS(tanggal_lahir, 6) AS tambah_6_bulan,
    SYSDATE AS waktu_sistem,
    LAST_DAY(tanggal_lahir) AS akhir_bulan_lahir,
    NEXT_DAY(tanggal_lahir, 'FRIDAY') AS jumat_berikutnya,
    FLOOR(MONTHS_BETWEEN(SYSDATE, tanggal_lahir)/12) AS usia_dalam_bulan,
    ROUND(tanggal_lahir, 'MONTH') AS pembulatan_bulan
FROM mahasiswa;
```

Screenshoot Output :

	FORMAT_HARI	FORMAT_TAHUN	JAM_SEKARANG	TAMBAH_6_BULAN	WAKTU_SISTEM	AKHIR_BULAN_LAHIR	JUMAT_BERIKUTNYA	USIA_DALAM_BULAN	PEMBULATAN_BULAN
1	15 / SAT	2004 / 04 / 04	21:26:47 PM	15-11-2004	05-03-2026	31-05-2004	21-05-2004	21	01-05-2004
2	20 / THU	2003 / 03 / 03	21:26:47 PM	20-05-2004	05-03-2026	30-11-2003	21-11-2003	22	01-12-2003
3	10 / MON	2005 / 05 / 05	21:26:47 PM	10-07-2005	05-03-2026	31-01-2005	14-01-2005	21	01-01-2005
4	25 / WED	2004 / 04 / 04	21:26:47 PM	25-02-2005	05-03-2026	31-08-2004	27-08-2004	21	01-09-2004
5	05 / FRI	2004 / 04 / 04	21:26:47 PM	05-09-2004	05-03-2026	31-03-2004	12-03-2004	22	01-03-2004
6	12 / FRI	2003 / 03 / 03	21:26:47 PM	12-06-2004	05-03-2026	31-12-2003	19-12-2003	22	01-12-2003
7	18 / SAT	2005 / 05 / 05	21:26:47 PM	18-12-2005	05-03-2026	30-06-2005	24-06-2005	20	01-07-2005
8	30 / SAT	2004 / 04 / 04	21:26:47 PM	30-04-2005	05-03-2026	31-10-2004	05-11-2004	21	01-11-2004
9	14 / SAT	2004 / 04 / 04	21:26:47 PM	14-08-2004	05-03-2026	29-02-2004	20-02-2004	22	01-02-2004
10	09 / TUE	2003 / 03 / 03	21:26:47 PM	09-03-2004	05-03-2026	30-09-2003	12-09-2003	22	01-09-2003
11	07 / WED	2004 / 04 / 04	21:26:47 PM	07-01-2005	05-03-2026	31-07-2004	09-07-2004	21	01-07-2004
12	01 / FRI	2005 / 05 / 05	21:26:47 PM	01-10-2005	05-03-2026	30-04-2005	08-04-2005	20	01-04-2005
13	11 / THU	2004 / 04 / 04	21:26:47 PM	11-05-2005	05-03-2026	30-11-2004	12-11-2004	21	01-11-2004
14	28 / MON	2005 / 05 / 05	21:26:47 PM	31-08-2005	05-03-2026	28-02-2005	04-03-2005	21	01-03-2005
15	05 / MON	2003 / 03 / 03	21:26:47 PM	05-11-2003	05-03-2026	31-05-2003	09-05-2003	22	01-05-2003

c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char)) :

Script Table :

```
SELECT
TO_NUMBER('175') + 5 AS tes_tambah,
TO_CHAR(tanggal_lahir, 'Day, DD Month YYYY') AS tgl_rapi,
TO_CHAR(berat_badan, '999.99') AS berat_format,
TO_DATE('05-03-2026', 'DD-MM-YYYY') AS string_ke_tgl
FROM mahasiswa
```

Screenshoot output :

	TES_TAMBAH	TGL_RAPI	BERAT_FORMAT	STRING_KE_TGL
1	180	Saturday , 15 May 2004	72.00	05-03-2026
2	180	Thursday , 20 November 2003	65.00	05-03-2026
3	180	Monday , 10 January 2005	50.00	05-03-2026
4	180	Wednesday, 25 August 2004	80.00	05-03-2026
5	180	Friday , 05 March 2004	48.00	05-03-2026
6	180	Friday , 12 December 2003	68.00	05-03-2026
7	180	Saturday , 18 June 2005	52.00	05-03-2026
8	180	Saturday , 30 October 2004	85.00	05-03-2026
9	180	Saturday , 14 February 2004	55.00	05-03-2026
10	180	Tuesday , 09 September 2003	70.00	05-03-2026
11	180	Wednesday, 07 July 2004	66.00	05-03-2026
12	180	Friday , 01 April 2005	45.00	05-03-2026
13	180	Thursday , 11 November 2004	62.00	05-03-2026
14	180	Monday , 28 February 2005	50.00	05-03-2026
15	180	Monday , 05 May 2003	88.00	05-03-2026

d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n)) :

Script Table :

```
SELECT nama,berat_badan,
ABS(-72) AS contoh_abs,
MOD(id, 2) AS sisa_bagi_id,
FLOOR(berat_badan) AS bulatan_bawah,
POWER(id, 2) AS id_pangkat,
ROUND(berat_badan, 0) AS bulatan_terdekat,
CEIL(berat_badan) AS bulatan_atas
FROM mahasiswa;
```

Screenshoot output :

	NAMA	BERAT_BADAN	CONTOH_ABS	SISA_BAGI_ID	BULATAN_BAWAH	ID_PANGKAT	BULATAN_TERDEKAT	BULATAN_ATAS
1	Ulung Pratama	72	72	1	72	1	72	72
2	Budi Santoso	65	72	0	65	4	65	65
3	Siti Aminah	50	72	1	50	9	50	50
4	Andi Wijaya	80	72	0	80	16	80	80
5	Dewi Lestari	48	72	1	48	25	48	48
6	Fajar Ramadhan	68	72	0	68	36	68	68
7	Gita Gutawa	52	72	1	52	49	52	52
8	Hendra Gunawan	85	72	0	85	64	85	85
9	Indah Permata	55	72	1	55	81	55	55
10	Joko Widodo	70	72	0	70	100	70	70
11	Kevin Sanjaya	66	72	1	66	121	66	66
12	Lesti Kejora	45	72	0	45	144	45	45
13	Maulana Malik	62	72	1	62	169	62	62
14	Nina Bobo	50	72	0	50	196	50	50
15	Oscar Pratama	88	72	1	88	225	88	88